Associative property

Any binary operation $*$ defined on a nonempty set S is said to satisfy the associative property, if

$$a*(b*c)=(a*b)*c \quad \forall a,b,c \in S .$$

Existence of identity property

An element $e \in S$ is said to be the **Identity Element** of S under the binary operation $*$ if for all $a \in S$ we have that $a*e=a$ and $e*a=a$.

Existence of inverse property

If an identity element e exists and if for every $a \in S$, there exists b in S such that $a*b=e$ and $b*a=e$ then $b \in S$ is said to be the **Inverse Element** of a. In such instances, we write $b=a^{-1}$.

Note

a^{-1} is an element of S. It should be read as the inverse of a and not as $\frac{1}{a}$.

Note

(i) The **multiplicative identity** is 1 in $\mathbb{Z}$ and it is the one and only one element with the property $n \cdot 1 = 1 \cdot n = n, \ \forall n \in \mathbb{Z}$.

(ii) The **multiplicative inverse** of any element, say 2 in $\mathbb{Q}$ is $\frac{1}{2}$ and no other nonzero rational number x has the property that $2 \cdot x = x \cdot 2 = 1$.

Note

Whenever a mathematical statement involves 'for every' or ' for all' , it has to be proved for every pair or three elements. It is not easy to prove for every pair or three elements. But these types of definitions may be used to prove the negation of the statement. That is, negation of "for every" or "for all" is "there exists not". So, produce one such pair or three elements to establish the negation of the statement.

The questions of existence and uniqueness of identity and inverse are to be examined. The following theorems prove these results in the more general form.

Theorem 12.1: (Uniqueness of Identity)

In an algebraic structure the identity element (if exists) must be unique.

Proof

Let $(S,*)$ be an algebraic structure. Assume that the identity element of S exists in S.

It is to be proved that the identity element is unique. Suppose that e_1 and e_2 be any two identity elements of S.

First treat e_1 as the identity and e_2 as an arbitrary element of S.

Then by the existence of identity property, $e_2 * e_1 = e_1 * e_2 = e_2$. ... (1)

Interchanging the role of e_1 and e_2, $e_1 * e_2 = e_2 * e_1 = e_1$. ...(2)

From (1) and (2), $e_1 = e_2$. Hence the identity element is unique which completes the proof.

Theorem 12.2 (Uniqueness of Inverse)

In an algebraic structure the inverse of an element (if exists) must be unique.

Proof

Let $(S,*)$ be an algebraic structure and $a \in S$. Assume that the inverse of a exists in S. It is to be proved that the inverse of a is unique. The existence of inverse in S ensures the existence of the identity element e in S.

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